UNIT - II

SET THEORY

Basic concepts of set theory :- A set is any well-defined collection of objects called the elements (or) members of the set. A collection of well-defined objects is called a set. Each of the objects in the set is called a member of an element of the set.

Ex: Books, cities, numbers, animals etc.

Elements of a set are usually denoted by lower-case letters while sets are denoted by capital letters

Ex: Natural Numbers $N = \{1, 2, 3, 4, \ldots\}$

The owels $O = \{a, e, i, o, u\}$

Formalization of sets :- There are two ways by which the sets can be described.

1. **Roster Notation :-** Roster Notation is a complete list of all the elements of the set.

Ex: $B = \{2, 4, 6, 8, \ldots 40\}$

2. **Set-Builder Notation :-** It is used the roster method is impossible.

Ex: $B = \{ x / 2 \leq x \leq 40, \text{ and } 'x' \text{ is Even} \}$.

The vertical bar '/' is read as such that.

Symbol	Meaning
$\in$	Belongs to
$\forall$	for all
$\exists$	There Exist

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Sub set :- If every element in a set A is also element of a set B then A is called subset of B. It is denoted by $A \subseteq B$.

Note : The null set is a subset of every set and every set is a subset of itself.

ie $\phi \subseteq A$ and $A \subseteq A$.

Ex: If $A = \{1,3,4,5,8,9\}$, $B = \{1,2,3,5,7\}$, $C = \{1,5\}$

Then $C \subseteq A$ and $C \subseteq B$.

proper subset :- If A is a subset of B and 'B' is not subset of A then A is called a proper subset of B it is denoted by $A \subset B$.

If A is subset of B then 'B' is called a super set of A and it is denoted by $B \supset A$.

Note : Two different sets are Equal ie $A = B$ if and only if $A \subset B$ and $B \subset A$.

Types of set :-

1. **Null set :-** The set with no elements is called an Empty set (or) Null set. A null set is denoted by ϕ. The null set is a subset of every set ie $\phi \subset A$

2. **countable set :-** A countable set is a set with the same cardinality as some subset of the set of natural Numbers.

Ex: $R = \{ x/x \text{ is a natural number} \}$

$\Rightarrow$ 'R' is an infinite set

$A = \{ x/x \text{ is a Even number between '2' and 40} \}$

$\Rightarrow$ A is a finite set.

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3. **Uncountable set** :- The uncountability of a set is closely related to its cardinal Number, a set is uncountable if its cardinal number is larger than that of the natural numbers.

4. **Universal set** :- In many discussions all the sets are considered to be subsets of one particular set. This set is called the Universal set for that discussion. The universal set is denoted by 'μ' (or) 'U'

5. **power set** :- The set of all subsets of a set 'A' is called the power set of A. and it is denoted by P(A). If A has 'n' elements in it then P(A) has 2^n elements.

Ex: If $A = \{1, 2, 3\}$ then

$P(A) = \{\phi, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$

6. **Equivalent sets** :- Two sets are said to be Equivalent if there is a one-one correspondence between the elements of the two sets.

Ex: If $A = \{a, b, c\}$, $B = \{\alpha, \beta, \gamma\}$ then 'A' and 'B' are Equivalent sets.

7. **Disjoint sets** :- Two sets are said to be disjoint if they have no element in common.

Ex: If $A = \{1, 2, 3\}$, $B = \{6, 7, 8, 9\}$ then 'A' and 'B' are disjoint sets. Here $A \cap B = \phi$.

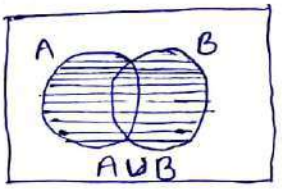
Union of two sets :- The union of two sets 'A' and 'B' is the set whose elements are all of the elements "in A" or "in B" or "in both". The Union of sets 'A' and 'B'

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is denoted by $A \cup B$ is read as "A union 'B".

Ex: Let $A = \{1, 2, 3, 4\}$ $B = \{3, 4, 5, 6, 7\}$

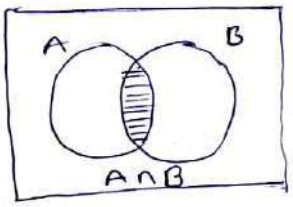
$A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$.



Intersection of two sets :- The intersection of two sets 'A' and 'B' is the set whose elements are all of the elements common to both 'A' and 'B'. The intersection of the sets of 'A' and 'B' is denoted by $A \cap B$ is read as "A intersection B".

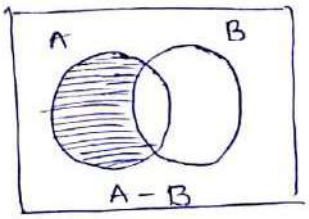
Ex: Let $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6, 7\}$

$A \cap B = \{3, 4\}$.



Difference of sets :- If A and B are subsets of the universal set μ then the complement of B in A is the set of all elements in A which are not in A. and it is denoted by $A - B$

$$A - B = \{ x / x \in A \text{ and } x \notin B \}.$$

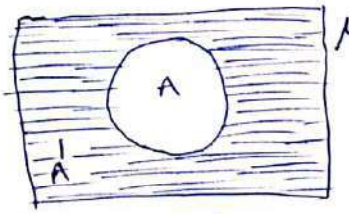


Ex: If $A = \{1, 2, 3, 4, 5\}$, $B = \{2, 3, 4\}$

$A - B = \{1, 2, 3, 4, 5\} - \{2, 3, 4\} = \{1, 5\}$

Complement of a set :- If 'μ' is a universal set containing the set A then $\mu - A$ is called the complement of A and it is denoted by A'.

$$\mu - A = A' = \{ x / x \notin A \}.$$



Ex: Let $\mu = \{1, 2, 3, 4, 5, \ldots\}$, $A = \{1, 2, 3, 4\}$

$A' = \mu - A = \{5, 6, 7, 8, \ldots\}$. $B = \{2, 4, 6, 8\}$

$B' = \mu - B = \{1, 3, 5, 7, 9, 10 \ldots\}$.

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Inclusion - Exclusion principle :- The inclusion-Exclusion principle is a counting technique which generalizes the familiar method of obtaining the number of elements in the union of two finite sets. Symbolically expressed as

$$n(A \cup B) = n(A) + n(B) - n(A \cap B).$$

where $n(A)$ = number of elements in A

$n(B)$ = number of elements in B.

$n(A \cup B)$ = no. of elements both 'A' and 'B'

$n(A \cap B)$ = no. of elements in A and B with common.

For any finite sets A, B, C we have

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C).$$

Ex: Find the no. of mathematics students at a college taking at least one of the languages Telugu, Hindi and English given the following data.

65 study Telugu, 20 study Telugu and Hindi
45 study Hindi, 25 study Telugu and English
42 study English, 15 study Hindi and English.
8 study all three Languages.

Sol: Given $n(T) = 65$, $n(H) = 45$, $n(E) = 42$

$n(T \cap H) = 20$, $n(T \cap E) = 25$, $n(H \cap E) = 15$,

$n(T \cap H \cap E) = 8$

By inclusion - Exclusion principle

$$n(T \cup H \cup E) = n(T) + n(H) + n(E) - n(T \cap H) - n(T \cap E) - n(H \cap E) + n(T \cap H \cap E)$$

$$= 65 + 45 + 42 - 20 - 25 - 15 + 8$$

$$= 100 \text{ //}$$

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Cartesian product :- cartesian product is a direct product of sets. Let us consider two sets A and B the set of all order pair (a,b) where $a \in A$ and $b \in B$ is called the product of A and B and is denoted by $A \times B$.

Ex: Let $A = \{1,2,3\}$ $B = \{4,5\}$ then

$A \times B = \{(1,4), (1,5), (2,4), (2,5), (3,4), (3,5)\}$

$B \times A = \{(4,1) (5,1), (4,2), (5,2), (4,3), (5,3)\}$

$\therefore A \times B \neq B \times A$.

Laws of set theory :-

1. Identity Laws : $A \cup \phi = A$ and $A \cap \mu = A$
2. Domination Laws: $A \cup \mu = \mu$ and $A \cap \phi = \phi$
3. Idempotent Laws : $A \cup A = A$ and $A \cap A = A$
4. Complementation Law : $(A')' = A$
5. Commutative Laws : $A \cup B = B \cup A$ and $A \cap B = B \cap A$
6. Associative Laws : $A \cup (B \cup C) = (A \cup B) \cup C$

   $A \cap (B \cap C) = (A \cap B) \cap C$
7. Distributive Laws : $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

   $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
8. DeMorgan's Laws : $(A \cup B)' = A' \cap B'$

   $(A \cap B)' = A' \cup B'$.
9. Absorption Laws : $A \cup (A \cap B) = A$ and $A \cap (A \cup B) = A$
10. Complement Laws : $A \cup A' = \mu$ and $A \cap A' = \phi$.

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Ex: prove that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Proof: Let 'x' is an arbitrary element of $A \cup (B \cap C)$

Then $x \in A \cup (B \cap C) \iff x \in A$ (or) $x \in (B \cap C)$

$\iff x \in A$ or ($x \in B$ and $x \in C$)

$\iff (x \in A$ or $x \in B)$ and $(x \in A$ or $x \in C)$

$\iff x \in (A \cup B)$ and $x \in (A \cup C)$

$\iff x \in (A \cup B) \cap (A \cup C)$

Hence $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Ex: prove that $(A \cap B)' = A' \cup B'$

Proof: Let $x \in (A \cap B)'$

$\iff x \notin A \cap B$

$\iff \sim (x \in A \cap B) \iff \sim (x \in A$ and $x \in B)$

$\iff \sim (x \in A)$ or $\sim (x \in B)$

$\iff x \notin A$ or $x \notin B \iff x \in A'$ or $x \in B'$

$\iff x \in A' \cup B'$

Hence $(A \cap B)' = A' \cup B'$.

Ex: prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ by using member-ship table.

Proof:

A	B	C	B∪C	A∩(B∪C)	A∩B	A∩C	(A∩B)∪(A∩C)
Y	Y	Y	Y	Y	Y	Y	Y
Y	Y	N	Y	Y	Y	N	Y
Y	N	Y	Y	Y	N	Y	Y
Y	N	N	N	N	N	N	N
N	Y	Y	Y	N	N	N	N
N	Y	N	Y	N	N	N	N
N	N	Y	Y	N	N	N	N
N	N	N	N	N	N	N	N

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Relations :- Any set of order pairs defines a binary relation. and it is arbitrary association of elements with in a set or with elements of another set.

Mathematically Let A and B are any two sets. A relation from A to B is a subset of $A \times B$.

Ex: $A = \{a, b\}$ and $B = \{a\}$

$A \times B = \{(a,a), (b,a)\}$

Any subset of $A \times B$ is called relation, ie $(a,a) \in A \times B$

(a,a) is a relation.

Note: 1. $(a,b) \in R$ then we say "a is related to b" and it can be written as "aRb" (or) $R(a,b)$

2. $(a,b) \notin R$ then we say "a is not related to b" and it can be written as "$a \not R b$".

ordered pair :- If $a \in B$ and $b \in B$ then the ordered pair is the set $\{\{a\}, \{a,b\}\}$ consisting of the pair $\{a,b\}$ and the singleton $\{a\}$. It is represented by (a,b).

In the order pair (a,b), the element 'a' is called the first element and 'b' is called the second element.

If (a,b) and (a',b') are two ordered pairs then.

$(a,b) = (a',b') \Rightarrow a = a'$ and $b = b'$.

Note: 1. (a,b) and (b,a) are two different ordered pairs

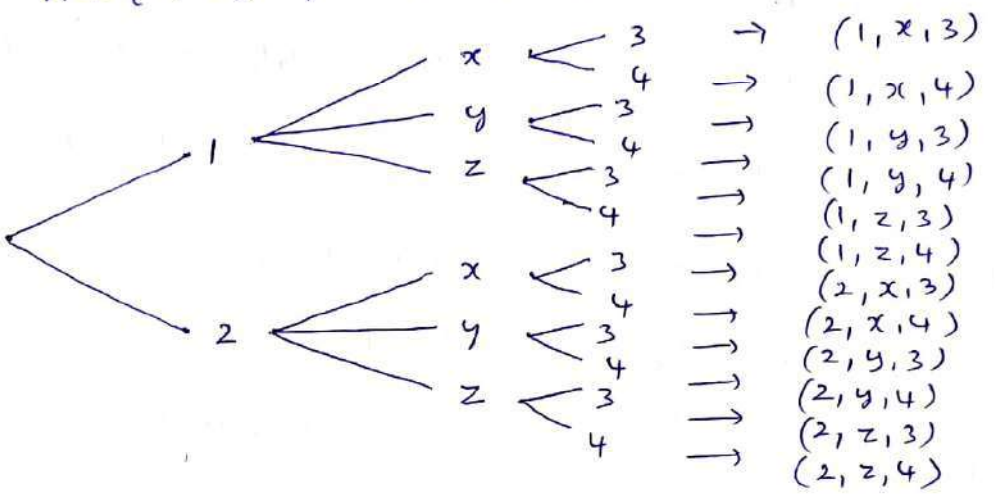
$(a,b) \neq (b,a)$.

2. The two objects in an ordered pairs do not have to be distinct. ie (a,a) is a well-defined ordered pair.

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Ex: Let $A = \{1, 2\}$, $B = \{x, y, z\}$, $C = \{3, 4\}$ then find $A \times B \times C$.

Sol: $A = \{1, 2\}$, $B = \{x, y, z\}$, $C = \{3, 4\}$



Domain and Range of relations :-

The domain of a relation R is the set of all first elements of the ordered pair which belongs to R and the Range of 'R' is the set of second element.

1. Domain (R) $= \{ a \in A / aRb \text{ for some } b \in B \}$

2. Range (R) $= \{ b \in B / aRb \text{ for some } a \in A \}$.

Ex: If $A = \{1, 2, 3\}$, $B = \{a, b, c\}$ then find domain and Range. If $R = \{(1, a), (1, c), (2, b), (3, a)\}$.

Sol: Given $A = \{1, 2, 3\}$, $B = \{a, b, c\}$

$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c), (3, a), (3, b), (3, c)\}$

Given Relation $R = \{(1, a), (1, c), (2, b), (3, a)\}$

Domain R = The set of first elements in R

$= \{1, 2, 3\}$

Range R = The set of second elements in R

$= \{a, c, b\}$.

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properties of Binary Relations :-

1. **Universal relation :-** A relation R on a set A is Universal if $R = A \times A$.

Ex: If $A = \{1,2,3\}$ then $A \times A = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$ is the universal relation on A.

2. **void relation :-** A relation R on a set A is void relation if R is the null set. ϕ.

Ex: If $A = \{3,4,5\}$ then $R = \{a+b > 10\}$ is a null set

3. **Identity relation :-** A relation R on a set A is Identity relation If $R = \{(a,a) / a \in A\}$ and it is denoted by I_A.

Ex: If $A = \{1,2,3\}$ then $R = \{(1,1), (2,2), (3,3)\}$.

4. **Inverse relation :-** If 'R' is any relation from a set 'A' to set 'B' the inverse of R is denoted by R^{-1}.

$$R^{-1} = \{(y,x) / y \in B, x \in A, (x,y) \in R\}$$

Ex: If $A = \{2,3,5\}$, $B = \{6,8,10\}$ then $R = \{(2,6), (2,8), (2,10), (3,6), (5,10)\}$

Now $R^{-1} = \{(6,2), (8,2), (10,2), (6,3), (10,5)\}$

5. **Reflexive relation :-** A relation on a set A is reflexive if aRa for every $a \in A$ i.e If $(a,a) \in R$, $\forall x \in a$

Ex: If $A = \{1,2,3,4\}$ then $R = \{(1,1), (1,2), (2,1), (2,2), (3,3), (4,4)\}$

$\therefore$ 'R' is called reflexive relation. otherwise it is called Irreflexive. Ex: $R = \{(1,1), (2,2), (2,3), (4,4)\}$.

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6. Symmetric relation :- A relation R on a set A is called symmetric if $(b,a) \in R$ whenever $(a,b) \in R$ for all $a,b \in A$.

7. Anti symmetric relation :- R is an anti symmetric if there exists (a,b) and $(b,a) \in R$ then $a = b$.

8. Transitive relation :- A relation R on a set A is transitive if whenever $a R b$ and $b R c$ then $a R c$. ie $(a,b), (b,c) \in R$ then $(a,c) \in R$.

EX: If $A = \{1,2,3,4\}$ and $R = \{(1,1)\ (1,3), (2,4), (3,1)\ (3,3)\ (4,3)\}$ then find if relation is reflexive, symmetric and transitive.

Sol: $R = \{(1,1), (1,3), (2,4), (3,1), (3,3), (4,3)\}$

i) Reflexive : $R \cup \{(2,2), (4,4)\}$

$\therefore R = \{(1,1), (1,3), (2,4), (3,1), (3,3), (4,3), (2,2), (4,4)\}$

ii) symmetric :- $R \cup \{(4,2), (3,4)\}$

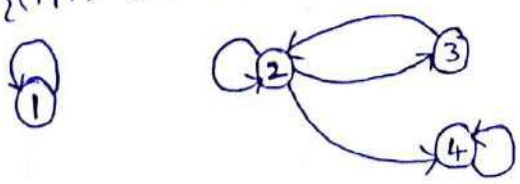
$\therefore R = \{(1,1), (1,3), (2,4), (3,1), (3,3), (4,3), (4,2), (3,4)\}$

iii) Transitive :- $R \cup \{(2,3), (4,1)\}$

$\therefore R = (1,1), (1,3), (2,4), (3,1), (3,3), (4,3), (2,3), (4,1)\}$

EX: If $A = \{1,2,3,4\}$ and $R = (1,1), (2,2), (2,3), (3,2), (4,2)\ (4,4)\}$ then draw its directed graph.

Sol: $R = \{(1,1)\ (2,2)\ (2,3)\ (3,2), (4,2), (4,4)\}$



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Matrix representation of relations :-

Let A and B are both finite sets and 'R' is a relation from A to B. Then R may be represented as a matrix called the relation matrix of R. It is denoted by M_R.

Let $A = \{a_1, a_2, \dots a_m\}$ and $B = \{b_1, b_2 \dots b_n\}$ are two finite sets containing 'm' and 'n' elements respectively.

$M_R = [m_{ij}]_{m \times n}$ is defined by

$$m_{ij} = \begin{cases} 0 & \text{if } (a_i, b_j) \in R \\ 1 & \text{if } (a_i, b_j) \notin R. \end{cases}$$

Ex: Let $A = \{1,2,3,4\}$ and $R = \{(x,y) / x > y\}$ find M_R.

Sol: Given $A = \{1,2,3,4\}$

$R = \{(4,1)\ (4,3)\ (4,2)\ ,(3,1)(3,2)\ ,(2,1)\}$

The corresponding relation matrix for the R are given

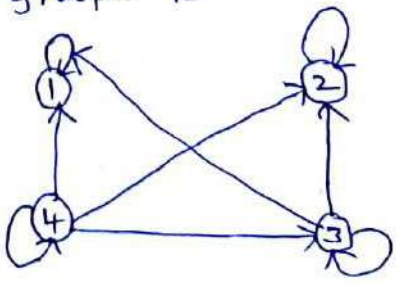
$$M_R = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{array}_{4 \times 4}.$$

Ex: If $A = \{1,2,3,4\}$ defined by $R = (1,1)\ (2,2)\ (3,3)$, $(4,4), (4,3), (4,2), (4,1), (3,2), (3,1)\}$. Find the matrix and directed graph of relation R.

Sol: The matrix of relation R is a 4x4 matrix is

$$M_R = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \end{array}.$$

directed graph is



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Equivalence relations :- A relation R in a set A is called an Equivalence relation if it is reflexive, symmetric and transitive.

Let a, b, c are arbitrary elements of some set A. Then "$a \sim b$" (or) "$a \equiv b$" denotes 'a' is Equivalent to 'b'.

i) Reflexive : $a \sim a$

ii) Symmetric : If $a \sim b$ then $b \sim a$

iii) Transitive : If $a \sim b$ and $b \sim c$ then $a \sim c$.

Ex: Let $A = \{1,2,3,4\}$ and $R = \{(1,1), (1,4), (4,1) (4,4) (2,2) (2,3), (3,2), (3,3)\}$. prove that 'R' is an Equivalence.

Sol: Given $R = \{(1,1) (1,4) (4,1) (4,4) (2,2) (2,3) (3,2) (3,3)\}$

i) Reflexive :- For any $x \in A \Rightarrow (x,x) \in R$

$\therefore xRx \Rightarrow$ 'R' is reflexive. $\left\{ \because \{(1,1) (2,2) (3,3) (4,4)\} \right.$

ii) Symmetric :- For any $x, y \in A \Rightarrow$ If $(x,y) \in R$ then $(y,x) \in R$

$\therefore xRy \Rightarrow yRx \Rightarrow$ 'R' is symmetric. $\left\{ \because (1,4) \to (4,1) \text{ and } (2,3) \to (3,2) \right.$

iii) Transitive :- For any $x, y, z \in A$

If (x,y) and $(y,z) \in R$ then $(x,z) \in R$

$\therefore xRy$ and $yRz \Rightarrow xRz$. 'R' is transitive.

Hence the relation 'R' is an Equivalence Relation.

$\left\{ \because \begin{array}{l} (1,1) ; (1,4) \to (1,4) \\ (2,3), (3,2) \to (2,2) \\ (3,2) (2,3) \to (3,3) \\ \text{etc...} \end{array} \right.$

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Congruence Relation :- Congruence relation is an example of equivalence relation. Let 'a' and 'b' are any two integers and 'r' be a fixed positive integer then 'a' and 'b' are said to be congruent modulo 'r' ie Mathematically $a \equiv b \pmod r$ if 'r' devides $(a-b)$.

Ex: $83 \equiv 13 \pmod 5$

since $83 - 13 = 70$ is divisible by '5'

Ex: Prove that the relation "congruence modulo m" over the set of positive integers is an equivalence relation

Sol: Let $x, y \in N$, $x \equiv y \pmod m$ if and only if $x-y$ is divisible by 'm'.

i) Let $x, y, z \in N$ then

$x - x = 0 \cdot m \Rightarrow x \equiv x \pmod m$ for all $x \in N$

ii) Let $x \equiv y \pmod m$. Then $x-y$ is divisible by 'm'

$\Rightarrow -(x-y) = y-x$ is also divisible by 'm'

ie $y \equiv x \pmod m$

$\therefore$ The relation $\equiv$ is symmetric.

iii) Let $x \equiv y \pmod m$ and $y \equiv z \pmod m$

ie $(x-y)$ and $(y-z)$ are divisible by 'm'

Now $(x+y)+(y-z)$ is divisible by 'm'.

ie $x-z$ is divisible by 'm'

$\Rightarrow x \equiv z \pmod m$ $\therefore$ The relation $\equiv$ is transitive

Hence '$\equiv$' is equivalence relation.

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Ex: Let 'R' denote a relation on the set of order pairs of positive integers such that $(x,y)\,R\,(u,v)$ iff $xv = yu$ show that R is an Equivalence relation.

Sol: Let x, y, u and v be positive integers

Given $(x,y)\,R\,(u,v) \Leftrightarrow xv = yu$

i) since $xy = yx$ is true for all positive integers

$\Rightarrow (x,y)\,R\,(x,y)$

$\therefore$ The reation R is reflexive.

ii) Let $(x,y)\,R\,(u,v) \Rightarrow xv = yu$

$\Rightarrow yu = xv \Rightarrow uy = vx$

$\Rightarrow (u,v)\,R\,(x,y)$

$\therefore$ The relation 'R' is symmetric.

iii) Let x, y, u, v, m and 'n' are positive integers

Let $(x,y)\,R\,(u,v)$ and $(u,v)\,R(m,n)$

$\Rightarrow xv = yu$ and $un = vm$

$\Rightarrow xvun = yu\,vm \Rightarrow xn = ym$ $\because$ by cancelling uv

$\Rightarrow (x,y)\,R\,(m,n)$.

$\therefore$ The relation 'R' is transitive.

Hence The relation 'R' is Equivalence relation.

Ex: N is the set of Natural numbers. The relation 'R' is defined by $(a,b)\,R\,(c,d) \Leftrightarrow a+d = b+c$. Prove that R is an Equivalence relation.

Sol:

Let a, b, c and d be natural numbers.

Given $(a,b)\,R\,(c,d) \Leftrightarrow a+d = b+c$.

i) $(a,b)\,R\,(a,b) \Rightarrow a+b = b+a$

$\therefore$ R is reflexive.

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ii) $(a,b) R (c,d) \Rightarrow a+d = b+c$

$\Rightarrow b+c = a+d$

$\Rightarrow c+b = d+a \Rightarrow (c,d) R (a,b)$

$\therefore$ 'R' is symmetric

iii) $(a,b) R (c,d)$ and $(c,d) R (m,n)$

$\Rightarrow a+d = b+c$ and $c+n = d+m$

$\Rightarrow a+d+c+n = b+c+d+m$

$\Rightarrow a+n = b+m \Rightarrow (a,b) R (m,n)$

$\therefore$ 'R' is transitive

Hence 'R' is an Equivalence relation.

Compatibility relation :- A Relation 'R' in X is said to be a compatibility relation if it is reflexive and symmetric. so all Equivalence relations are compatibility relations and it is sometimes denoted by "$\simeq$".

Composition of relations :- Let 'R' be a relation from x to y and 'S' be a relation from y to z. Then a relation written as R o S is called a composite relation of 'R' and 'S' where $R \circ S = \{ (x,z) / x \in X, z \in Z$ there exist $y \in Y$ with $(x,y) \in R$ and $(y,z) \in S \}$.

Ex: Let $R = \{(1,2) (3,4) (2,2) \}$ and $S = \{ (4,2), (2,5), (3,1) (1,3) \}$. Find R o S and S o R.

Sol: Given $R = \{(1,2), (3,4), (2,2) \}$

$S = \{ (4,2), (2,5), (3,1), (1,3) \}$

$\therefore R \circ S = \{ (1,5), (3,2), (2,5) \}$

$S \circ R = \{ (4,2), (3,2), (1,4) \}$ clearly $R \circ S \neq S \circ R$.

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partial ordering :- A binary relation 'R' in P is called a partial order relation (or) a partial ordering in P iff R is reflexive, antisymmetric and transitive. ie

1. aRa for all $a \in P$
2. aRb and $bRa \Rightarrow a=b$
3. aRb and $bRc \Rightarrow aRc$

Ex: show that the relation "greater than or equal to" is a partial ordering on the set of integers.

sol: Let 'z' be the set of all integers and the relation $R = "\geq"$.

i) since $a \geq a$ for every integer a,
the relation '$\geq$' is reflexive

ii) Let 'a' and 'b' be any two integers
Let aRb and $bRa \Rightarrow a \geq b$ and $b \geq a \Rightarrow a=b$
$\therefore$ The relation "$\geq$" is antisymmetric.

iii) Let a, b and 'c' be any three integers.
Let aRb and $bRc \Rightarrow a \geq b$ and $b \geq c \Rightarrow a \geq c$
$\therefore$ The relation '$\geq$' is transitive.

Hence the relation '$\geq$' is partial ordering on the set of integers $\therefore (\mathbb{Z}, \geq)$ is a poset.

poset (partially ordered set) :- A set 'P' together with a partial ordering 'R' is called a partial ordered set or poset. The relation R is often denoted by the symbol '$\leq$' then the ordered pair $(P, \leq)$ is called poset.

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Hasse diagrams :- A partial order $\leq$ on a set P can be represented by means of a diagram known as Hasse diagram of $(P, \leq)$.

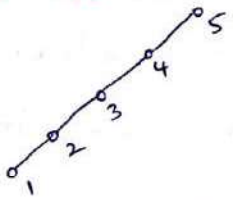
i) Each element is represented by a small circle or dot

ii) The circle for $x \in P$ is drawn below the circle for $y \in P$ if $x < y$ and a line is drawn between x and y if y covers 'x'.

iii) If $x < y$ but y does not cover 'x' then x and y are not connected directly by a single line.

Ex: Let $P = \{1, 2, 3, 4, 5\}$ and $\leq$ be the relation "Less than or Equal to" then the Hasse diagram is drawn below.

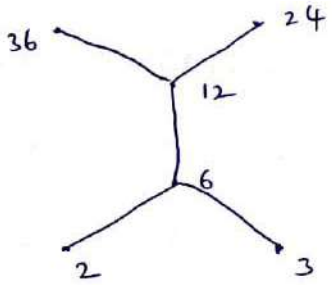
Sol:



It is totally order set.

Ex: Let $X = \{2, 3, 6, 12, 24, 36\}$ and the relation $\leq$ be such that $x \leq y$ if x divides y. Draw the Hasse diagram of $(X, \leq)$.

Sol:



it is not a total order set.

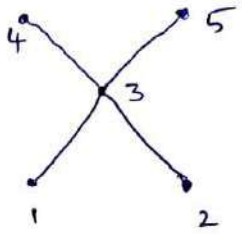
Ex: Draw the Hasse diagram for the relation R on A = $\{1, 2, 3, 4, 5\}$ whose relation matrix given below.

$$M_R = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

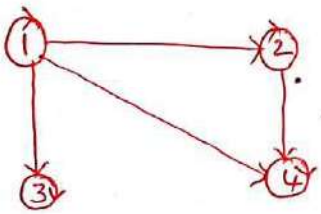
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Sol: $R = \{ (1,1), (1,3), (1,4), (1,5), (2,2), (2,3), (2,4), (2,5), (3,3), (3,4), (3,5), (4,4), (5,5) \}$.

Hasse diagram for M_R is



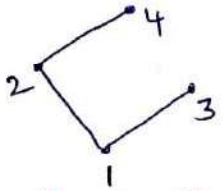
Ex: A partial order 'R' on the set $A = \{1,2,3,4\}$ is represented by the following digraph. Draw the Hasse diagram for R.



Sol: By examining the given digraph we find that

$R = \{ (1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4) \}$.

∴ Hasse diagram is

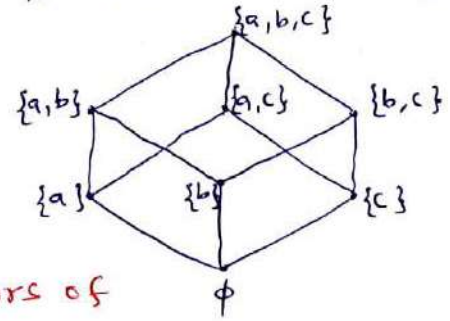


Ex: Draw the Hasse diagram for the partial ordering $\subseteq$ on the power set $P(S)$ where $S = \{a,b,c\}$

Sol: Given $S = \{a,b,c\}$

$P(S) = \{ \phi, \{a\}, \{b\}, \{c\}, \{a,b\}, \{b,c\}, \{a,c\}, \{a,b,c\} \}$.

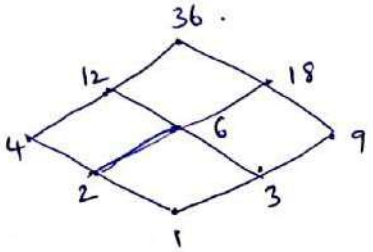
∴ Hasse diagram is



Ex: Draw the Hasse diagram representing the positive divisors of 36.

Sol: $D_{36} = \{ 1,2,3,4,6,9,12,18,36 \}$

∴ Hasse diagram is



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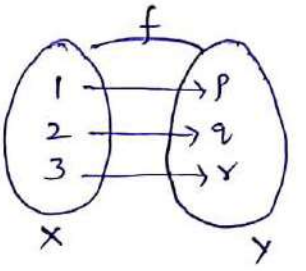
Functions :-

Let X and Y be any two sets. A relation 'f' from 'X' to 'Y' is called a function if for every $x \in X$ there is a unique element $y \in Y$ such that $(x, y) \in f$.

Ex: Let $X = \{1, 2, 3\}$, $Y = \{p, q, r\}$ and $f = \{(1, p), (2, q), (3, r)\}$

Then $f(1) = p$, $f(2) = q$, $f(3) = r$

clearly 'f' is a function from $X \to Y$



Domain and range of a function :-

If $f : X \to Y$ is a function then 'x' is called the Domain of 'f' and the set 'y' is called the codomain of f. The range of 'f' is defined as the set of all images under 'f'. It is denoted by $f(x) = \{ y / \text{ for some } x \text{ in } X, f(x) = y \}$

Ex: If the function 'f' is defined by $f(x) = x^2 + 1$ on the set $\{-2, -1, 0, 1, 2\}$ find the range of 'f'.

Sol: Given $f(x) = x^2 + 1$

$f(-2) = (-2)^2 + 1 = 5$, $f(0) = 0 + 1 = 1$, $f(2) = 4 + 1 = 5$

$f(-1) = (-1)^2 + 1 = 2$, $f(1) = 1 + 1 = 2$,

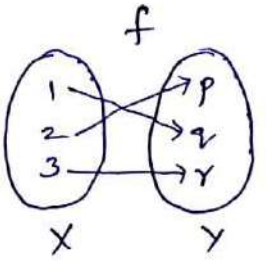
$\therefore$ The range of 'f' $= \{1, 2, 5\}$.

Types of functions :-

1. **one-to-one function (Injection) :-** A mapping $f : X \to Y$ is called one-to-one if distinct elements of 'x' are mapped into distinct elements of y

ie f is one-to-one if

$f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ for $x_1, x_2 \in X$



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Ex: $f: R \to R$ defined by $f(x) = 3x$, $\forall x \in R$ is one-one, since

$$f(x_1) = f(x_2) \Rightarrow 3x_1 = 3x_2 \Rightarrow x_1 = x_2, \forall x_1, x_2 \in R.$$

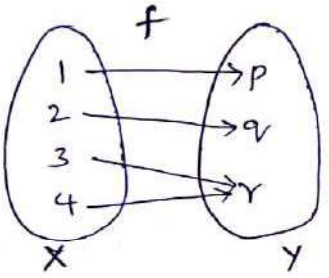
2. onto function (Surjection) :- A mapping $f: X \to Y$ is called onto if the range set $R_f = Y$.

If $f: X \to Y$ is onto, then each element of Y is f-image of atleast one element of X.

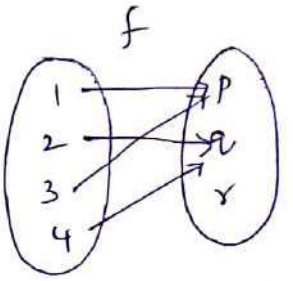
Mathematically $\forall y \in B$, $\exists x \in A$ such that $f(x) = y$.

Note: If 'f' is not onto then it is said to be into.

Ex:



Surjective



not surjective.

3. Bijection function :- A mapping $f: X \to Y$ is called Bijective function if it is one-to-one and onto. Such a mapping is also called a one-to-one correspondence between 'X' and 'Y'.

Ex: Show that a mapping $f: R \to R$ defined by $f(x) = 2x+1$ for $x \in R$ is a bijective map from R to R.

Sol: Let $f: R \to R$ defined by $f(x) = 2x+1$, for $x \in R$.

i) one-to-one : Let 'x' and 'y' be any two elements in R such that $f(x) = f(y)$

$$\Rightarrow 2x+1 = 2y+1$$

$$\Rightarrow 2x = 2y \Rightarrow x = y$$

$\therefore f(x) = f(y) \Rightarrow x = y$

$\therefore$ f is one-to-one.

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ii) onto :- Let 'y' be any element in the domain R

$\Rightarrow f(x) = y \Rightarrow 2x+1 = y \Rightarrow x = \frac{(y-1)}{2}$

clearly $x = \frac{(y-1)}{2} \in R$. $\therefore f$ is onto

Hence 'f' is a bijective map.

Composition of Functions:- Let $f: X \to Y$ and $g: Y \to Z$ be two functions. Then the composition of 'f' and 'g' denoted by gof is the function from 'X' to 'Z' defined as $(gof)(x) = g(f(x))$. for all $x \in X$.

Ex:- Let $f(x) = x+2$, $g(x) = x-2$, and $h(x) = 3x$ for $x \in R$ where R is the set of real numbers. Find gof, fog, fof, gog, foh, hog, hof and fogoh.

Sol: Given $f(x) = x+2$, $g(x) = x-2$, $h(x) = 3x$

i) $(gof)(x) = g(f(x)) = g(x+2) = x+2-2 = x$

ii) $(fog)(x) = f(g(x)) = f(x-2) = x-2+2 = x$

iii) $(fof)(x) = f(f(x)) = f(x+2) = x+2+2 = x+4$

iv) $(gog)(x) = g(g(x)) = g(x-2) = x-2-2 = x-4$

v) $(foh)(x) = f(h(x)) = f(3x) = 3x+2 = 3x+2$

vi) $(hog)(x) = h(g(x)) = h(x-2) = 3(x-2) = 3x-6$

vii) $(hof)(x) = h(f(x)) = h(x+2) = 3(x+2) = 3x+6$.

viii) $(fogoh)(x) = f((goh)(x)) = f(g(h(x))) = f(g(3x))$

$= f(3x-2) = 3x-2+2 = 3x$ //

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Ex: If $f(x) = x^2+3x+1$ and $g(x) = 2x-3$ then find the composition of i) fof ii) fog iii) gof.

Sol: . . . . . .

Inverse functions :- A function $f: X \to Y$ is said to be invertible of its inverse function f^{-1} is also function from the range of 'f' into 'x'.

If $f(x) = y$ then $x = f^{-1}(y)$.

Ex: Let $f: R \to R$ be given by $f(x) = x^3 - 2$. Find f^{-1}.

Sol: i) Let $x_1, x_2 \in R$ such that $f(x_1) = f(x_2)$

$\Rightarrow x_1^3 - 2 = x_2^3 - 2 \Rightarrow x_1^3 = x_2^3 \Rightarrow x_1 = x_2$

$\therefore$ f is one-to-one.

ii) If $f(x) = y$ then $y = x^3 - 2 \Rightarrow x^3 = y+2$

$\Rightarrow x = \sqrt[3]{y+2}$ $\therefore$ 'f' is onto.

$\Rightarrow f^{-1}$ is invertible and $f^{-1}(x) = \sqrt[3]{x+2}$.

Ex: $f: R \to R$ is defined by $f(x) = ax+b$ for $a, b \in R$. show that 'f' is invertible and find the inverse of 'f'.

Sol: 'f' is invertible and $f^{-1}(x) = \frac{x-b}{a}$

Floor and ceiling functions :- Let 'x' be a real number, then the least integer that is not less than 'x' is called the ceiling of x. The ceiling of x is denoted by $\lceil x \rceil$.

Ex: $\lceil 2.15 \rceil = 3$, $\lceil \sqrt{5} \rceil = 3$, $\lceil -7.4 \rceil = -7$, $\lceil -2 \rceil = -2$

Let 'x' be a real number, then the greatest integer that does not exceed 'x' is called the 'Floor' of 'x' The Floor of x is denoted by $\lfloor x \rfloor$.

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EX: $\lfloor 5.14 \rfloor = 5$ $\lfloor \sqrt{5} \rfloor = 2$, $\lfloor -7.6 \rfloor = 8$

Recursive function :- A recursive function is a function which is defined in terms of itself. We can also define functions recursively.

A recursive definition has two parts.

1. Definition of the smallest argument ie $f(0)$ or $f(1)$
2. Definition of $f(n)$, given $f(n-1)$, $f(n-2)$ etc...

EX: An example of a recursively defined function.

$f(0) = 3$, $f(n+1) = 2f(n) + 3$. Then

$f(1) = 2f(0) + 3 = 2 \cdot (3) + 3 = 9$

$f(2) = 2f(1) + 3 = 2 \cdot (9) + 3 = 21$

$f(3) = 2f(2) + 3 = 2(21) + 3 = 45$

$f(4) = 2f(3) + 3 = 2(45) + 3 = 93 \cdots\cdots$

EX: calculate the values of the following recursively defined function $f(0) = 1$ and $f(n) = n f(n-1)$.

Sol: Given $f(0) = 1$ and $f(n) = n f(n-1)$

$f(1) = 1 \cdot f(0) = 1 \cdot 1 = 1$

$f(2) = 2 \cdot f(1) = 2 \cdot 1 = 2$

$f(3) = 3 \cdot f(2) = 3 \cdot 2 = 6$

$f(4) = 4 \cdot f(3) = 4 \cdot 6 = 24 \cdots\cdots$

This is the recursive definition of the factorial function, $f(n) = n!$.

EX: $f(1) = 1$, $f(2) = 1$ and $f(n) = f(n-2) + f(n-1)$

EX: $f(0) = 0$, $f(n) = f(n-1) + 2n - 1$

EX: $f(0) = 5$, $f(n) = f(n-1) + 2$.

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EX: Consider the following recursive function

If $x < y$ then $f(x,y) = 0$, if $y \leq x$ then $f(x,y) = f(x-y, y)+1$

Find the value of $f(4,7)$ and $f(19,6)$.

Sol: Given $f(x,y) = \begin{cases} 0, & x < y \\ f(x-y, y)+1, & y \leq x. \end{cases}$

i) $f(4,7) = 0$.

ii) $f(19,6) = f(19-6, 6)+1 = f(13,6)+1$

$f(13,6) = f(13-6, 6)+1 = f(7,6)+1$

$f(7,6) = f(7-6, 6)+1 = f(1,6)+1 = 0+1 = 1$

Now $f(13,6) = f(7,6)+1 = 1+1 = 2$.

$f(19,6) = f(13,6)+1 = 2+1 = 3$ //.

LATTICES :- In this section, we introduce Lattices which have important applications in the theory and design of computers.

Def :- A Lattice is a partially ordered set $(L, \leq)$ in which every subset consisting of pair of elements has a lowest upper bound and greatest lower bound.

A Lattice 'L' is an algebraic system with binary operations 'v' and '$\wedge$' It is denoted by $(L, \wedge, v)$. Thus 'L' is called Lattice if the following axioms hold where $a, b, c \in L$.

i) Commutative Law : $(a \wedge b) = (b \wedge a)$, $(a v b) = (b v a)$

ii) Associative Law : $a \wedge (b \wedge c) = (a \wedge b) \wedge c$,
$a v (b v c) = (a v b) v c$

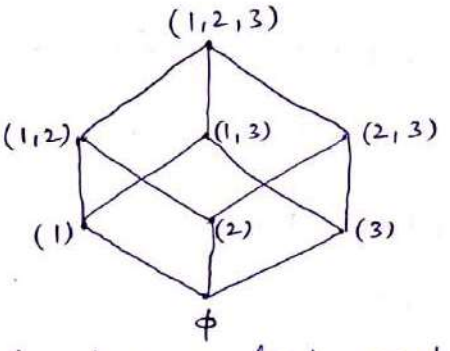
iii) Absorption Law : $a \wedge (a v b) = a$, $a v (a \wedge b) = a$.

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The LUB is denoted by $(a \vee b)$ and is called "join" of 'a' and 'b'. The 'GLB' is denoted by $(a \wedge b)$ and is called "meet" of 'a' and 'b'.

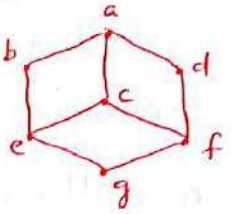
EX: Let $X = \{1,2,3\}$ then the poset $(P(X), \subseteq)$ of all Sub sets of X is a Lattice or not

Sol: $P(X) = \{\phi, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$



Each subset has a lub and glb. So it is Lattice.

EX: Determine whether the following Hasse diagram represent Lattice or not.



Sol:

LUB :

∨	a	b	c	d	e	f	g
a	a	a	a	a	a	a	a
b	a	b	a	a	b	a	b
c	a	a	c	a	c	c	c
d	a	a	a	d	a	d	d
e	a	b	c	a	e	c	e
f	a	a	c	d	c	f	f
g	a	b	c	d	e	f	g

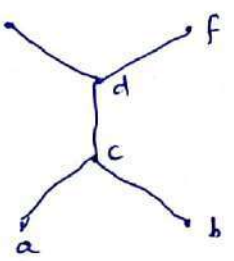
GLB :

∧	a	b	c	d	e	f	g
a	a	b	c	d	e	f	g
b	b	b	e	g	e	g	g
c	c	e	c	f	e	f	g
d	d	g	f	d	g	f	g
e	e	e	e	g	e	g	g
f	f	g	f	f	g	f	g
g	g	g	g	g	g	g	g

Each subset of two elements has a least upper bound and a greatest lower bound so it is a lattice.

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Ex: The following diagram is a not a lattice because GLB (a,b) and LUB (e,f) does not exist.



Some properties of Lattices :-

If 'L' is any Lattice then $\forall\ a,b,c \in L$

i) $a \wedge a = a$, $a \vee a = a$ — Idempotent

ii) $a \wedge b = b \wedge a$, $a \vee b = b \vee a$ — Commutative.

iii) $a \wedge (b \wedge c) = (a \wedge b) \wedge c$
$a \vee (b \vee c) = (a \vee b) \vee c$ — Associative.

iv) $a \wedge (a \vee b) = a$
$a \vee (a \wedge b) = a$ — Absorption.

v) $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$
$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$ — Distributive.

Duality Law in Lattice :- The dual of any statement in a Lattice $(L, \wedge, \vee)$ is defined to be the statement that is obtained by interchanging '$\wedge$' and '$\vee$'

Ex: The dual of $a \wedge (b \vee c) = a \vee (b \wedge c)$

vi) $a \leq b$, $c \leq d \Leftrightarrow a \wedge c \leq b \wedge d$, $a \vee c \leq b \vee d$.

vii) $a \wedge b \leq a$, $b \leq a \vee b$

Ex: Let $(L, \leq)$ be a Lattice in which '$\wedge$' and '$\vee$' denote the operations of meet and join respectively. For any $a \in L$, $a \leq b \Leftrightarrow a \wedge b = a \Leftrightarrow a \vee b = b$.

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Sol: i) we shall first prove that $a \le b \Leftrightarrow a \wedge b = a$

Let us assume that $a \le b$, Also we know that $a \le a$

$\Rightarrow a \le a \wedge b$ {$\because$ From $a \wedge b \le a$

Hence $a \le b \Rightarrow a \wedge b = a$

Next assume that $a \wedge b = a$, But it is only possible if $a \le b$, that is $a \wedge b = a \Rightarrow a \le b$.

Finally $a \le b \Rightarrow a \wedge b = a$ and

$a \wedge b = a \Rightarrow a \le b$

Hence $a \le b \Leftrightarrow a \wedge b = a$.

Similarly we can prove that $a \le b \Leftrightarrow a \vee b = b$.

Ex: Let $(L, \le)$ be a Lattice Then $b \le c \Rightarrow \begin{cases} a \wedge b \le a \wedge c \\ a \vee b \le a \vee c \end{cases}$

Sol: By above Theorem $a \le b \Leftrightarrow a \wedge b = a \Leftrightarrow a \vee b = b$.

Now we have to show that $b \le c \Rightarrow a \wedge b \le a \wedge c$

we shall show that $(a \wedge b) \wedge (a \wedge c) = a \wedge b$

$\Rightarrow (a \wedge b) \wedge (a \wedge c) = a \wedge (b \wedge a) \wedge c$

$= a \wedge (a \wedge b) \wedge c$

$= (a \wedge a) \wedge (b \wedge c)$

$= (a \wedge a) \wedge (b \wedge c)$

$= a \wedge (b \wedge c)$

$= a \wedge b$

$\therefore$ If $b \le c$ then $a \wedge b \le a \wedge c$.

Similarly we can prove that

If $b \le c$ then $a \vee b \le a \vee c$.

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Some Special Lattices :-

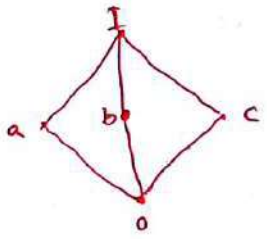
1. **Distributive Lattices :-** A Lattice $(L, \wedge, \vee)$ is distributive if the following additional identity holds for all x, y and z in L.

$$x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z) \quad \text{and}$$

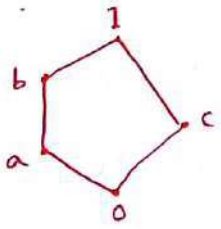
$$x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$$

Ex: Show that the following simple But significant Lattices are not distributive.

i)



ii)



Sol: i) the diamond Lattice is not distributive.

Now $a \wedge (b \vee c) = a \wedge I = a$ but

$(a \wedge b) \vee (a \wedge c) = 0 \vee 0 = 0$

$\therefore a \wedge (b \vee c) \neq (a \wedge b) \vee (a \wedge c)$.

ii) The pentagon Lattice is also not distributive.

for these three elements ..

$a \wedge (b \vee c) = a \wedge (I) = a \qquad a \neq 0$.

$(a \wedge b) \vee (a \wedge c) = a \vee 0 = 0$

Ex: Consider the poset $(L, \leq)$ where $L = \{1, 2, 3, 5, 30\}$ and partial ordered relation $\leq$ is defined as if x and $y \in L$ then $x \leq y$ means 'x divides y' Then show that poset is a Lattice.

Sol: since $GLB(x, y) = x \wedge y = lcm(x, y)$

$LUB(x, y) = x \vee y = gcd(x, y)$

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Now we can construct the operation table I and II for GLB and LUB respectively and the Hasse diagram is shown in Fig.

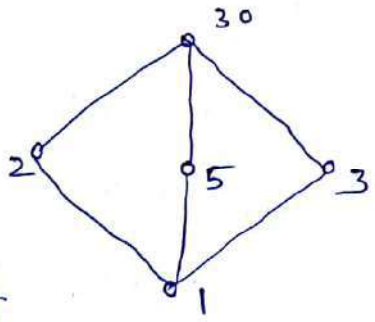


Table I :

LUB	1	2	3	5	30
1	1	2	3	5	30
2	2	2	30	30	30
3	3	30	3	30	30
5	5	30	30	5	30
30	30	30	30	30	30

Table II

GLB	1	2	3	5	30
1	1	1	1	1	1
2	1	2	1	1	2
3	1	1	3	1	3
5	1	1	1	5	5
30	1	2	3	5	30

Now Test for distributive Lattice.

Assume $x = 2$, $y = 3$, $z = 5$ then

$x \wedge (y \vee z) = 2 \wedge (3 \vee 5) = 2 \wedge 30 = 2$

$(x \wedge y) \vee (x \wedge z) = (2 \wedge 3) \vee (3 \wedge 5) = 1 \vee 1 = 1$

since LHS $\neq$ RHS

Hence Lattice is not a distributive Lattice.

2. **Bounded Lattice :-** A bounded Lattice is an algebraic system $(L, \wedge, \vee)$ if there exist a greatest element '1' and a least element '0' in the Lattice i.e

1) for all $x \in L$, $x \wedge 1 = x$ and $x \vee 1 = 1$

2) for all $x \in L$, $x \wedge 0 = 0$ and $x \vee 0 = x$

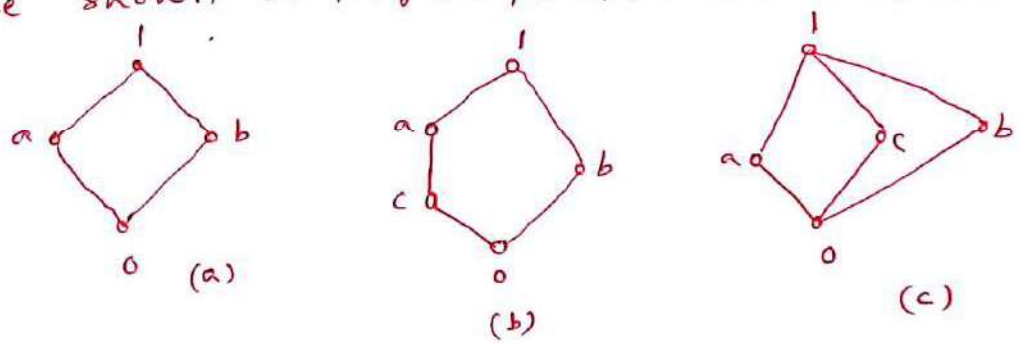
The element '1' is called the upper bound (or) top of L and the element '0' is called the Lower bound (or) bottom of L.

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3 Complemented Lattice :- A Lattice $(L, \wedge, \vee)$ is said to complemented Lattice if, Every element in the Lattice has a complement. (or)

A bounded Lattice with greatest element '1' and least element '0' then for the elements $a, b \in L$ element 'b' is complement of a if $a \vee b = 1$ and $a \wedge b = 0$.

Ex: Lattice shown in Fig (a), (b), (c) are complemented Lattices



Sol:

a) $GLB(a,b) = a \wedge b = 0$

$LUB(a,b) = a \vee b = 1$

Hence it is complemented Lattice

b) $GLB(a,b) = a \wedge b = 0$, $GLB(c,b) = c \wedge d = 0$

$LUB(a,b) = a \vee b = 1$ $LUB(c,b) = c \vee b = 1$

So both 'a' and 'c' are Complements of 'b'.

Hence it is complemented Lattice.

c) $GLB(a,c) = a \wedge c = 0$, $GLB(a,b) = a \wedge b = 0$

$LUB(a,b) = a \vee b = 1$, $LUB(a,c) = a \vee c = 1$.

So both 'b' and 'c' are Complements of 'a'.

Hence it is complemented Lattice :

4. Modular Lattice :- A modular Lattice is a Lattice that satisfies the following self-dual condition.

$x \leq b \Rightarrow x \vee (a \wedge b) = (x \vee a) \wedge b$.

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Ex: In a distributive Lattice $(L, \leq)$, $a \wedge b = a \wedge c$ and $a \vee b = a \vee c$ imply that $b = c$

Sol: Given $a \wedge b = a \wedge c$ and $a \vee b = a \vee c$

Now $b = b \wedge (a \vee b)$ $\quad \because$ absorption Law

$= b \wedge (a \vee c)$ $\quad \because a \vee b = a \vee c$

$= (b \wedge a) \vee (b \wedge c)$

$= (a \wedge b) \vee (b \wedge c)$

$= (a \wedge c) \vee (b \wedge c)$ $\quad \because a \wedge b = a \wedge c$

$= (c \wedge a) \vee (c \wedge b)$

$= c \wedge (a \vee b)$

$= c \wedge (a \vee c)$

$b = c$ $\quad \because$ by absorption

//

Ex: In a distributive Lattice If an element has a complement then this complement is Unique.

Sol: Suppose that an element a has two complements 'b' and 'c'. Then $a \vee b = 1$ $\quad a \wedge b = 0$

$a \vee c = 1$ $\quad a \wedge c = 0$

We have $b = b \wedge 1$

$= b \wedge (a \vee c)$ $\quad \because a \vee c = 1$

$= (b \wedge a) \vee (b \wedge c)$

$= 0 \vee (b \wedge c)$

$= (a \wedge c) \vee (b \wedge c)$

$= (a \vee b) \wedge c$

$= 1 \wedge c$

$b = c$

//

Ex: Every distributive Lattice is Modular.

Sol: Let $(L, \leq)$ be a distributive Lattice if $a \leq c \Rightarrow a \vee c = c$

Now $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c) = (a \vee b) \wedge c$

$\therefore$ Hence Every distributive Lattice is modular.

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